

KAESONG, Korea, Democratic People's Republic of

WMO: 470700

Lat: 37.967N

Lon: 126.567E

Elev: 70

StdP: 100.49

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-11.0	-22.8	0.5	-8.2	-20.1	0.6	-7.8	10.1	-9.0	9.0	-6.0	1.7	320	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	31.6	25.8	30.2	24.7	28.9	23.9	26.9	30.3	26.1	28.8	25.4	27.7	2.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.5	28.5	25.3	20.6	27.8	24.7	19.9	27.1	85.1	30.5	81.5	28.8	78.6	27.8	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.0	6.7	-15.7	33.8	2.5	1.5	-17.5	34.9	-19.0	35.8	-20.4	36.6	-22.2	37.7		
(5)			-16.3	28.3	2.4	1.6	-18.0	29.4	-19.4	30.4	-20.7	31.3	-22.4	32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-3.4	-0.9	4.7	11.2	17.0	21.8	24.6	25.3	20.8	14.3	6.3	-1.1	(6)
(7)		DBStd	10.53	3.94	3.83	3.47	3.52	2.67	2.17	2.14	2.30	2.54	3.46	4.41	4.21	(7)
(8)		HDD10.0	1391	416	305	167	27	0	0	0	0	8	125	343		(8)
(9)		HDD18.3	3011	674	538	421	216	59	2	0	8	129	361	602		(9)
(10)		CDD10.0	2043	0	0	4	62	217	353	453	476	323	141	14	0	(10)
(11)		CDD18.3	622	0	0	0	1	17	106	195	218	81	4	0	0	(11)
(12)		CDH23.3	4142	0	0	0	9	119	557	1293	1769	378	17	0	0	(12)
(13)		CDH26.7	1016	0	0	0	1	14	112	323	521	45	1	0	0	(13)
(14)	Wind	WSAvg	2.0	1.8	2.1	2.5	2.3	2.1	2.0	2.1	1.8	1.7	1.8	1.7		(14)
(15)	Precipitation	PrecAvg	1350	15	27	36	68	108	137	409	290	137	52	46	21	(15)
(16)		PrecMax	2290	55	71	125	168	272	535	1009	835	557	206	112	65	(16)
(17)		PrecMin	599	1	0	3	10	11	27	140	34	5	3	9	2	(17)
(18)		PrecStd	368	13	20	27	35	61	109	197	197	129	50	25	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	11.9	17.9	24.7	28.4	31.0	33.7	33.6	29.3	25.4	18.7	10.1	(19)
(20)			MCWB	4.5	7.5	10.8	14.6	18.7	22.0	26.3	26.9	22.0	18.0	14.3	6.8	(20)
(21)		2%	DB	5.4	8.5	14.5	21.5	26.1	29.2	31.2	32.2	28.0	23.1	16.2	8.3	(21)
(22)			MCWB	2.6	4.4	9.1	13.9	17.8	21.3	25.5	26.4	21.8	16.8	12.4	5.8	(22)
(23)		5%	DB	3.7	6.5	12.4	19.4	24.4	27.9	29.7	30.8	26.8	21.5	14.5	6.7	(23)
(24)			MCWB	1.3	3.2	7.6	12.7	17.2	21.0	24.9	25.7	21.3	15.9	11.4	4.2	(24)
(25)		10%	DB	2.5	4.8	10.5	17.5	22.7	26.5	28.3	29.4	25.6	20.0	12.9	5.1	(25)
(26)			MCWB	0.3	2.0	6.3	11.7	16.5	20.5	24.2	25.1	20.4	14.8	10.2	2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	8.9	12.3	17.5	21.0	23.9	27.9	28.2	25.1	19.7	15.6	8.2	(27)
(28)			MCD B	6.9	10.7	15.7	21.2	25.4	28.1	31.4	31.4	27.2	23.1	17.5	9.5	(28)
(29)		2%	WB	3.3	5.3	10.1	15.3	19.3	23.0	26.6	27.2	23.7	18.1	13.6	6.1	(29)
(30)			MCD B	4.7	7.7	13.4	19.7	23.7	27.0	29.9	30.6	26.1	21.1	15.3	7.9	(30)
(31)		5%	WB	1.8	3.6	8.3	13.9	18.4	22.2	25.7	26.5	22.7	17.1	12.1	4.6	(31)
(32)			MCD B	3.3	6.0	11.6	18.2	22.5	25.7	28.4	29.6	25.1	20.2	13.9	6.3	(32)
(33)		10%	WB	0.5	2.3	6.8	12.6	17.4	21.5	25.1	25.9	21.7	16.1	10.5	3.1	(33)
(34)			MCD B	2.2	4.5	9.8	16.4	21.1	24.7	27.2	28.4	24.1	19.0	12.4	4.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.8	9.0	9.9	9.8	8.2	5.8	6.7	8.6	9.9	8.4	7.7	(35)
(36)			MCDBR	9.1	10.4	11.6	12.5	12.7	10.8	8.2	8.5	9.9	11.2	10.0	8.7	(36)
(37)			MCWBR	7.0	6.9	7.1	6.3	6.0	4.7	4.1	3.9	4.7	5.7	6.7	6.8	(37)
(38)		5% WB	MCD BR	8.0	9.4	10.5	10.7	10.3	8.6	6.7	7.1	7.8	9.5	8.5	8.2	(38)
(39)			MCWBR	7.0	7.2	7.1	6.4	5.9	4.4	3.9	3.8	4.7	5.7	6.8	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.420	0.480	0.557	0.610	0.621	0.695	0.667	0.573	0.504	0.501	0.478	0.423	(40)	
(41)		taud	1.939	1.840	1.719	1.655	1.681	1.598	1.707	1.923	2.027	1.911	1.908	1.963	(41)	
(42)		Ebn at Noon	707	715	703	697	701	649	663	718	745	688	639	669	(42)	
(43)		Edh at Noon	138	174	217	244	242	263	235	185	157	160	143	127	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.51	3.25	4.27	4.97	5.59	5.23	4.07	4.38	4.21	3.58	2.41	2.17	(44)	
(45)		RadStd	0.15	0.25	0.40	0.35	0.55	0.56	0.61	0.60	0.43	0.24	0.24	0.15	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	+57	

Nomenclature: See separate page